

Vampire-as-bite and vampire-as-mosquito-thief in the closed-vial engines

A note on the closed-vial vampire trees. Locks stay on the trees.

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Index of locks: <https://gist.github.com/martialsystems/12835f747d6360781f3cc7f91f243178>

Abstract

Two questions. First: under an explicit diet ladder and an allowed population crash, does a *Drosophila*-like sponging labellum evolve prestomal-tooth rasping and a costly bite that becomes the host meal, with F off 1? Second: can diploid sponging flies live on vertebrate blood already in an engorged mosquito, without a mammal-biting pierce kit?

Parent engine fly_vial @e2e22b7: k=3, seed 1, N=1,000, 80 generations, IBD F = 0.524 versus random F = 0.034. Census held. Courtship fell on both arms.

vial_sanguis @97aa7c8 (index still names @b7347ce): crash and exudate recovery are repeatable. Random flickers a biter and loses it. k-NN majority kit at F=1. Origin seed-1 knn: t_first_biter=211, census to 3 at t=6, recovered t=28, majority t=1,468, at t=2,500 n=1,200, F=1.000, p_biter=1. Kinship cap phi_max=0.25: two of three seeds, F about 0.24. Closing wounds and tears/sweat together extincts everyone. Engine retired.

vial_sanguis2 science 4946ab5 @7e32830: delayed cap plus pair fallback. One of three k-NN seeds held a kit at F about 0.23. That kit is exudate. Peak bite share 5.74% at t=4,644 with p_biter=1. Sweat plus wound at least 93.34% of host after hold. Vampire-as-bite fails as a 3-seed claim and as a meal share.

vial_morsus science b106228 @69f4f6d: decaying exudate bridge $\tau_{\text{bridge}}=400$. 0 of 3 k-NN. Recovered from crash, lived on the bridge, extinct when exudate hit 0 around t=400. Bite share never held at 0.50. Mid-bridge sweat plus wound about 0.98.

vial_culex science fa819c7 @ddb73a8: blank fruit founders. 0 of 3 k-NN. Extinct at t=5. Cargo still 200. Stolen share 0. They never found the mosquito.

vial_handoff science e843b8c @12d6952: living morsus k-NN at t=200 into the culex kitchen. 0 of 3. Hemolymph share 1.000. Stolen share at most 0.000067. gut_probe stayed untrained. Fail-predator.

Halt on every vampire tree.

Keywords

Drosophila melanogaster; closed vial; IBD; prestomal rasp; bite share; mosquito kleptoparasitism; exudate; delayed kinship cap.

1. Questions

Q1. Under an explicit diet ladder and an allowed population crash, does a *Drosophila*-like sponging labellum evolve prestomal-tooth rasping and then a costly bite, and how many generations / how deep a bottleneck does that take?

Q1b. Can that kit hold with F off 1 if kinship cap waits until after recovery and falls back to random pairing instead of vetoing every pair?

Q1c. After fruit is gone, does a prestomal rasp plus shallow bite become the majority of host calories on a decaying exudate bridge, with F off 1, on at least two of three seeds?

Q2. Can diploid sponging flies switch from fruit to stealing vertebrate blood already in an engorged mosquito, then find the next mosquito, without evolving a mammal-biting pierce kit?

Q2b. If living mid-bridge morsus k-NN flies (already past fruit-off, already rasping films) are put in the culex kitchen (finite engorged mosquitoes, no mammal skin, no exudate), can they add a gut probe and live on stolen midgut blood?

PASS on Q1c, Q2, Q2b is at least two of three k-NN seeds. 0 of 3 is a fail. Do not average.

2. Engine

Closed-form diploid QTL vial. Mating is k=3 nearest-neighbor on morphology plus courtship QTLs, or random. Hidden recessive load is excluded from similarity. F is founder-allele IBD, $1 - H_t/H_0$. Census ceiling 1,200 after the parent 1,000. Extinction is logged.

fly_vial @e2e22b7 is fruit only, 80 generations, N=1,000. Assortative F = 0.524 versus random F = 0.034 on seed 1. Wright random-mating form $1 - (1 - 1/2000)^{80} \approx 0.039$. Courtship t=1 to t=80: 0.934 to 0.595 assortative, 0.930 to 0.599 random. The courtship drop is not the assortative signature. Logs: logs/assort_80.json, logs/random_80.json. Do not restamp 0.524 / 0.034.

Vampire trees add a diet ladder. Fruit off at t_starve=5. Host channels: tears (k_tears=40), sweat, wound, bite. bite_weight=1.0. Standing costs on unused pierce, digest, heme_safe. phi_max=0.25 when the cap is on. z_max=3.0 from sanguis2 on. Fruit-forever is the negative control.

3. Vampire-as-bite

3.1 vial_sanguis (2026-09-16)

Six vials, k=3, N=1,000, 2,500 generations, t_starve=5.

seed	mate	min n	t_recover	flicker	t_held_biter	t_majority	final F	final p_biter
1	knn	3	28	211	1,108	1,468	1.000	1.000
1	random	1					extinct t=11	0
2	knn	18	22	412	638	1,211	1.000	0.914
2	random	6	16	162			0.821	0
3	knn	13	21	792	1,111	1,301	1.000	0.998
3	random	6	22	128			0.830	0

Fruit-forever to t=400: p_biter=0. Origin lock logs/vampire_2500_s1.json: t_first_biter=211.

Kinship cap phi_max=0.25: two of three seeds, F about 0.24 at 10k. Pierce still moved. Saliva did not. Seed 3 extinct at t=9.

Cap after recover: seed 1 extinct t=12 (all pairs vetoed). Seeds 2 and 3 held biters at F=0.238 / 0.236.

Host shift after hold: p_biter collapsed (0.005 / 0.007). Saliva left the negative region. Pierce did not keep rising.

Scab plus exudate-cap: closing wounds and tears/sweat together extincts both lines in one generation. Cliff. bite_weight stayed 1.0. Halt. Successor vial_sanguis2.

3.2 vial_sanguis2 (2026-09-16)

Delayed cap (n>=50 after starve, then phi_max=0.25). If the cap would accept fewer than 8 pairs, that generation uses random pairing.

seed	mate	min n	t_recover	flicker	t_held	F@1500	F@2500	final p_biter	fallback gens
1	fallback-dominated	2	28	128		0.887	0.939	0.001	2,490
1	random	7	19	137		0.680	0.805	0	0
2	knn delayed-cap	1						extinct t=11	0
2	random	3						extinct t=5	0
3	knn delayed-cap	11	21	109	754	0.228	0.236	0.936	1
3	random	23	16	142		0.578	0.760	0	0

Seed 1 is not a delayed-cap result. Seed 2 is bottleneck extinction. Seed 3 held a kit.

Host calories from logs/knn_delayed_2500_s3.json and logs/knn_delayed_10000_s3.json. Host calories: tears + sweat + usable_blood. Wound and bite split usable_blood in proportion to the logged channel means.

t	p_biter	F	tears %	sweat %	wound %	bite %	sweat+wound %
754 hold	0.058	0.236	1.32	52.20	45.42	1.06	97.62
2,500	0.936	0.236	0.95	57.25	39.49	2.31	96.74
4,644 peak bite	1.000	0.223	0.91	55.08	38.26	5.74	93.34
10,000	0.066	0.228	0.97	58.32	40.57	0.15	98.88

After t_held=754, sweat+wound is at least 93.34% of host. Peak bite share is 5.74%. p_biter counts flies with non-zero energy_bite on an exudate meal. Science 4946ab5. End of vial_sanguis2.

3.3 vial_morsus (2026-09-16)

The standing film is replaced by a decaying bridge ($\tau_{\text{bridge}}=400$ after first $n \geq 50$ post-starve). Then only bite is host food.

seed	mate	crash min n	t_recover	last live t	last n	last F	sw+wnd	bite share	extinct
1	knn	11	8	400	4	0.250	0.788	0	yes
2	knn	17	8	399	11	0.253	0.783	0.00016	yes
3	knn	38	6	398	8	0.264	0.770	0	yes
1	random	4	7	400	2	0.722	0.625	0.215	yes
2	random	4	7	400	16	0.521	0.801	0.00030	yes
3	random	12	7	398	8	0.542	0.787	0.000017	yes
1	fruit-forever			400	1,200	0.488	0	0.0000036	no

Mid-bridge (t_recover+200) every starve arm sat at n=1,200 with sweat+wound about 0.98. Rasp rose; pierce did not. When exudate_scale hit 0, census collapsed. 0 of 3 k-NN. Do not raise bite_weight. Science b106228. Halt. No 10k arm.

4. Vampire-as-mosquito-thief

4.1 vial_culex (2026-09-16)

Blank founders. Mosquito is the host. Stolen midgut blood and hemolymph are separate liquids. M0=200 at t_starve, arrivals a0=8 per generation.

seed	mate	last live t	last live n	last live F	t_end	cargo at t=5	stolen share	hemolymph share	extinct
1	knn	4	1,200	0.008	5	200	0	0	yes
2	knn	4	1,200	0.004	5	200	0	0	yes
3	knn	4	1,200	0.007	5	200	0	0	yes
1	random	4	1,200	0.001	5	200	0	0	yes

seed	mate	last live t	last live n	last live F	t_end	cargo at t=5	stolen share	hemolymph share	extinct
2	random	4	1,200	0.001	5	200	0	0	yes
3	random	4	1,200	0.001	5	200	0	0	yes
1	fruit-forever	400	1,200	0.551	400	0	0	0	no

None lived on hemolymph. None drained cargo. 0 of 3 k-NN. Science `fa819c7`. Halt. A hand-off of living morsus genomes is a different git.

4.2 vial_handoff (2026-09-16)

Mapped full living census at morsus $t=200$ ($n=1,200$, sweat+wound about 0.98). rasp copied to cuticle_rasp. $\text{find_mosquito} = 0.5 \text{ fluid_detect} + 0.5 \text{ seek}$. pierce not mapped. $\text{gut_probe} \sim N(0, 0.08)$. Exudate off. Hemolymph meals kill bodies.

seed	mate	F@200	rasp	find	n@2500	stolen	hemolymph	probe end	PASS
1	knn	0.235	1.000	0.343	1,200	0.000001	1.000	-0.475	no
2	knn	0.218	0.800	0.376	1,200	0	1.000	-0.386	no
3	knn	0.230	0.833	0.281	1,200	0.000067	1.000	-0.278	no
1	fruit-forever	0.235	1.000	0.343	1,200 at $t=400$	0.00081	0.111	-0.046	control
1	random	0.235	1.000	0.343	1,200	0.000011	1.000	-0.908	contrast

killed/gen=8 on every arm. map probe at snap is about 0. Film-rasp flies became mosquito predators (rasp $\times$ find). They did not add a gut probe. Fruit-forever stolen share $0.00081 < 0.05$. 0 of 3 k-NN. Science `e843b8c`. Halt. No 10k.

5. Repo table

Tree	SHA	Allowed sentence	Lock file
fly_vial	e2e22b7	k=3, seed 1, N=1,000, 80 generations: $F = 0.524$ vs random $F = 0.034$. Census held. Courtship fell on both arms.	logs/assort_80.json
vial_sanguis	97aa7c8	Crash and exudate recovery are repeatable. k-NN majority kit at $F=1$. Kinship cap two of three at F about 0.24. Exudate-cap is a cliff.	logs/vampire_2500_s1.json
vial_sanguis2	7e32830	Delayed-cap can hold F about 0.23. Peak bite share 5.74%. Sweat plus wound at least 93.34% of host. Science 4946ab5.	logs/knn_delayed_2500_s3.json
vial_morsus	69f4f6d	0 of 3 k-NN. Lived on the bridge, extinct at $t \sim 400$ when exudate hit 0. Science b106228.	logs/knn_2500_s1.json
vial_culex	ddb73a8	0 of 3 k-NN. Blank founders extinct at $t=5$; cargo still 200. Science fa819c7.	logs/knn_2500_s1.json
vial_handoff	12d6952	0 of 3. Fail-predator: lived on hemolymph. Science e843b8c.	logs/handoff_2500_s1.json

Index (pointers only): <https://gist.github.com/martialsystems/12835f747d6360781f3cc7f91f243178>

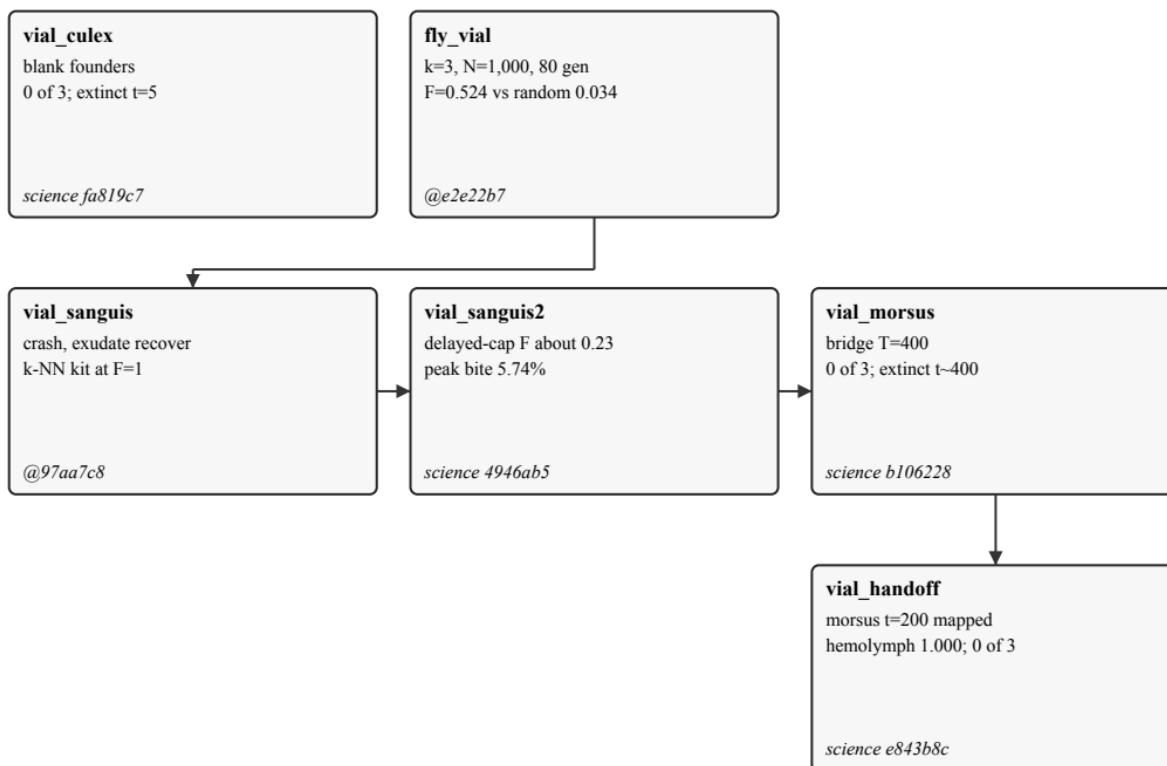
6. Replay

How to run: in each tree, `.venv/bin/python -m pytest`. Do not overwrite the listed lock files. Do not raise `bite_weight`. Do not raise `gut_probe` payoff. Do not restamp fly_vial F .

Tree	Lock
fly_vial	logs/assort_80.json, logs/random_80.json
vial_sanguis	logs/vampire_2500_s{1,2,3}.json, logs/random_2500_s{1,2,3}.json, logs/fruit_forever_400_s1.json
vial_sanguis2	logs/knn_delayed_2500_s{1,2,3}.json, logs/knn_delayed_10000_s3.json
vial_morsus	logs/knn_2500_s{1,2,3}.json

Tree	Lock
vial_culex	logs/knn_2500_s{1, 2, 3}.json
vial_handoff	logs/handoff_2500_s{1, 2, 3}.json; map maps/morsus_to_culex.json; t_snap=200

Figure 1. Parent F lock to bite arm to mosquito arm (2026-09-16).



Bite arm under fly_vial. culex is a sibling. handoff maps morsus into the culex kitchen.

Lineage of locked trees. Not a new measurement.

Revisions (2026-09-16)

2026-09-16: first note from the locked vampire trees. Numbers copied from the origin JSON cited in each README. Index gist 12835f74 left unedited.